PRODUCTION PLANNING

Version 3.5.2

Product Guide

FRONTISPIECE

Although every care has been taken in the production of this product information it cannot be guaranteed that the information contained herein is without error or omission.

This product information relates to: Production Planning 3.5.2

Contents

Glossary

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Active Production Order

This is a production order which has associated work-in-progress.

Activity Types

These are user definitions of activities to be reported and are linked to a JBA reporting type. There are system dependent activity types which are mandatory for the system to function; and user-defined activity types which may be defined to suit user requirements. The associated reporting type defines how the activity will affect updates to the database.

Actual Down Time

See Down Time.

AFI

Acronym for Advanced Financial Integrator.

Allocated Stock

The quantity of an item which has been allocated to customer orders or production orders or schedules. It is usually expressed as a balance at item/stockroom level.

Allocations

Reservation of inventory for consumption in a production order or schedule. It is a ‘soft’ reservation, and the material can be issued to any order, but it enables the application to calculate available quantities.

Amended Standard Production Orders

Production orders which are based on a standard route and only differ in detail.

Amortised Fixed Costs

This is the method of spreading fixed production costs over a designated batch size to ascertain the effect on unit product costs of the economies of scale production. See also Fixed Costs.

Archived Production Orders

Production orders which have been saved in an archive file and removed from the live order database. They are available for detailed enquiry.

Available

The quantity calculated by Planning to represent current availability on a given day equal to previous period available + supply – demand.

Available Stock

The quantity calculated by subtracting allocations from the physical stock balance. It represents uncommitted inventory which may be used to satisfy production demand.

Average Cost

This is a costing method employed by Inventory Management, whereby the weighted average unit cost of an item is recalculated every time a stock receipt is made.

Average Usage

The average usage per week/period of an item in a stockroom. The weeks or periods which are included in this calculation are defined by the usage profile.

Backflush

The automatic generation of standard material issues based on production quantities reported.

Backflush Item

An item that is designated to be automatically issued in production recording.

Backschedule

The calculation of operation and order start dates from the due date, using the lead time elements of the operations.

Balance

This may be used either to signify a database record holding summary information, such as a stockroom balance, or a single summary quantity field on such a record, such as allocated stock.

Base Version

The Version 3.x Production system is delivered in two editions; Base and Extended. The Base edition delivers functionality equivalent to that which was available in Version 2.0 The Extended edition provides additional function, notably scheduled (or repetitive) production and Process industry features such as co-products and potency.

Batch Balancing

This is a method of ensuring that the correct quantity and potency mix of materials is used in a production batch.

Bill of Material

The definition of the inputs that are required to make a product. Also known as a Product Structure, Recipe or Formula.

BOM

Acronym for Bill Of Material.

Booking

Work-in-progress reporting.

Booking History

A record of all material and production transactions posted during the progress of a production order or production schedule.

Bottleneck

This term is generally used to refer to a position on a production line where the production flow is constrained in some way. This can lead to build-ups of work and potentially have an adverse effect on the efficiency of a line or plant, and ultimately on profitability.

Bucket

In MPS and MRP, the period of time for which supply and demand are summarised for presentation.

Bucketless

This describes the MPS/MRP review process, which balances supply and demand on the date it is scheduled, rather than accumulating it into greater time periods.

Budget Capacity

The capacity of a work station that is compared with its load. It represents the capacity you expect to obtain from a work station. This can be 100% of stated capacity or a factor above or below 100% (see Standard Capacity)

By-Product

A product which is produced incidentally by a process which is primarily for the production of other products. It may have financial value, which will be deducted from the total costs of the mainstream product(s) and will also be treated as a negative cost, displayed in the Relief Cost element field.

Cancelled Production Order

A production order which has been aborted, and cannot be reopened.

Capacity

The amount of time that a work station is available for work in a given period.

Capacity Planning

The activity of calculating work station capacity requirements by comparison of duration for planned work with the capacity available for the planning period. The work schedule or the capacity may then be adjusted to obtain a balanced work flow.

Capacity Planning Run

The main function of the Capacity Requirements Planning application. This process calculates the work station capacity load that is required to achieve a particular production schedule according to scheduling rules.

Capacity Requirement

The time required at a work station by a particular piece of work or production schedule.

Cell

A group of stockrooms that are related for the purpose of material requirements planning.

Cellular Planning

A planning method by which the demand and supply of materials are identified and satisfied at cell level rather than model level.

Change Management

See Engineering Change Management.

Co-Products

Items that are necessarily produced together as a result of a production process. They share the burden of the cost of production.

Company Profile

A collection of control parameters specific to a Production company.

Completed Production Order

Production orders which have been completed. They cannot have bookings made against them. They may be reopened for further processing.

Component

Any item that is used in the production of another item. (See Input).

Component Location Reference

A method whereby components may be categorised by their location and position within an assembly, structure or process.

Confirmed Production Order

A production order with a firm commitment to produce an item, which cannot be changed in date or quantity except by explicit planner intervention.

Cost

A value associated with an item in a stockroom, or a movement. It is usually a value related to a single item (a unit cost), but may refer to a quantity of items (a movement cost or value).

Cost Apportionment Method

The method used to calculate the proportion of production costs that are applied to each item, when co-products are produced from a process.

Cost Centre

A functional or organisational area defined for the purposes of defining production costs. Each cost centre defines standard rates for labour, work station, set up and overheads. A cost centre is assigned to a work station and is used to calculate all standard production costs associated with that work station.

Cost Elements

16 cost elements are available to analyse costs. These are: relief costs, direct material, packaging, utility, labour, set up, machine, subcontract, overhead 1, overhead 2 (fixed), overhead 2 (variable), user-defined 1-4 and shrinkage.

Cost Relief Apportionment

The method used to calculate any By-Product Relief Costs that are applied to co-product costs in a co-product process.

Cost Roll Up

The method of generating product costs by calculating and accumulating costs of materials and operations required at each level of manufacture.

Costing Method

This refers to the method used to establish a cost for stock movements or stock balances. The methods available are latest, average, standard and FIFO (First In First Out).

Costing Route

The route designated for an item to calculate its unit cost within a stockroom. A unit cost may be calculated for each stockroom in which an item is stocked by designating a specific production route as a cost route.

Count Point

An operation at which WIP inventory is counted or reported.

Count Reporting Policy

This policy determines the method by which production quantities are recorded during booking. This may be total quantity or start and end quantity.

Creation Date

The date on which a production order is entered.

Crew Size

The standard number of operatives scheduled to work on an operation, either as direct labour or set up labour.

CRP

Acronym for Capacity Requirements Planning.

Cumulative Lead Time

The amount of time required to produce an item from scratch. It is based on a full explosion of the bills of material of the item and its sub-assemblies and includes the purchasing lead time of raw materials.

Current Cost

A category of cost. The application generates values for current and standard cost control. Current cost may be considered as the proposed standard cost for the next accounting period. See Standard Cost.

Current Date in Planning

The datum point of an MPS/MRP plan. The start date is determined by subtracting Overdue Days from this date. The Time Fence date is calculated from this date by adding the frozen Lead Time.

Customer Schedule

The forecast of a customer's expected delivery requirements. They can be at different status's in different time periods.

Customer Shelf Life

The amount of time an item must have left in its life when it is delivered to the customer. If an item is lot controlled, this time will be deducted from the Expiry Date to calculate the Last Available Date.

Delivery Area

Information which is used to identify where items should be moved to. Can be found on the Picking List.

Delivery Days Basis

This parameter is only pertinent to items which are not lot, batch or serial controlled. It allows delivery lead time to be taken into account during planning, and may be calculated using calendar days or working days. For lot controlled items, the Release Lead Time is used.

Delivery Lead Time

The delivery lead time value expressed in terms of the Delivery Days Basis.

Delivery Point

The exact position where items should be move to within the Delivery Area. Can be found on Picking List.

Demand

The forecast or actual requirement for an item.

Demand Policy

Policy which controls the comparison of sales forecasts with sales orders, customer schedules and dependent demand to arrive at the demand to drive MPS or MRP. This may be: no forecast; independent demand only; dependent and independent demand; dependent demand; explode forecasts to inputs; make to forecast only or total demand.

Dependent Demand

Demand for an item which is derived from the manufacture of a parent.

Descriptions File

This is a file maintained within the Inventory Management application which defines a number of parameter codes and their descriptions.

Discrete Manufacturing

A production control method where individual pieces of work are identifiable. Usually, production orders are used to manage this.

Down Time

The amount of time that a work station is out of action. The application provides the facility to record both planned and actual down time.

DRP

Acronym for Distribution Requirements Planning.

Duration Calculation Basis

The method by which the duration of an operation is calculated for scheduling purposes. This may be: set up time only; machine time + set up time; direct labour time + set up time; machine time + direct labour time + set up time; greater of machine time *or* direct labour time + set up time.

This can be set at Company Profile, Work Station or Route Operation level.

Economic Order Quantity

This is an optimum quantity of an item to be produced by a process route or supplied on an order. It may be entered for each process route and may be used as the basis of apportioning fixed costs for an item.

Effectivity

A method of controlling product input configurations. The effectivity of an input is the time period when it can be used in an assembly. The application uses an effective start date and an effective finish date to control input configurations. The system will ignore the item outside the effectivity dates.

Efficiency

The ratio of standard to actual performance.

Efficiency Variance

The difference between standard and actual performance in quantity and cost terms.

End Date (Planning)

The last date to be considered by the run. It can be entered or calculated as Current Date plus item cumulative lead time. It can be extended by setting a number of safety days

Engineering Change Management.

A system which controls and audits, via change requests, the addition, maintenance and deletion of items, route operations and input/outputs.

Equivalent Physical Quantity

This is used where item lots have variable potency. For an item lot with non-standard potency, it is the equivalent quantity of the item at standard potency. It is calculated as Physical Quantity x Actual Potency/Standard Potency

Exception Events

Transactions which are likely to cause a change in the supply and demand status of an item.

Expiry Date

The Expiry Date is calculated as lot Creation Date + Shelf Life

Extended Version

The Version 3.x Production system is delivered in two editions; Base and Extended. The Base edition delivers functionality equivalent to that which was available in Version 2.0 The Extended edition provides additional function, notably scheduled (or repetitive) production and Process industry features such as co-products and potency.

FIFO

This is an acronym for First In First Out - one of the costing methods available in the Inventory Management application. Using this method, each stock receipt is valued at actual cost, and issues are valued using these receipt batch costs on a First In First Out basis.

Filler Item

An item that is used to make up the required physical of a production batch, but which has no effect on the properties of the item produced. (See Balancing Quantity)

Finished Goods Receipt

The receipt of a quantity of a production item into an Inventory stockroom, as a result of a production order or schedule.

Firm Planned Production Order

A production order which remains under the control of the planner in terms of timing and quantity and is not recommended for change by Planning functions, unless Planning Filters are set to allow this.

Firming Period

The period for which firm work station schedules have been created.

First Available Date

For a lot controlled item, this is equal to the Creation Date + Release Lead Time (Days). It is the first date the item can be used.

Fixed Cost

An element of item cost that does not vary with the volume of production. Fixed elements of costs are; set up, fixed overhead, fixed user-defined costs.

Fixed Order Quantity

An ordering policy used by MPS and MRP to control suggested replenishment orders. It is used to generate suggested supplies of a pre-defined size.

Fixed Quantity Per

An input to a bill of material whose requirement will not vary with batch size.

Floor Stock

Inventory which is issued to a designated floor stock location (logical or physical stockroom) on the shop floor rather than directly for immediate consumption. Floor stock is consumed as it is used at a particular operation.

Floor Stock Location

This is a logical or physical stockroom where items with a Material Control Policy of issue to floor stock are issued and consumed.

Flow Route

This is a route where the individual operations are dependent on each other. Changes to schedules on flow routes for one operation result in changes to the whole route.

Formula

See Bill of Material.

Frozen Stock

The quantity of an item which is designated as ‘frozen’ and thus is not available for issue or allocation. It is expressed as a balance quantity at item/stockroom level, or item/lot level.

Generated Demand

See Dependent Demand.

Gross Requirement

The total demand for an item in a given time period before stock on-hand and supplies are netted.

GT Family

Group Technology code. This is a user-defined classification which may be used as a selection parameter both on a Selective MRP run and MPS and MRP reports.

Held Inventory Tracking

A regimen imposed by the system to force entry of a reference code/ description each time a WIP quantity is booked as ‘held’. This reference may be for the whole booked quantity or specific to one or more items in the total quantity. Any further movements of Held WIP Inventory (for example, transfer or scrap) necessitate the specification of the held inventory reference.

Held WIP Inventory

WIP inventory which is not available to progress to the next operation until released from held status. This may be because it is awaiting quality control inspection or rework.

In Transit

The quantity of an item which is currently in transit between two stockrooms. It is expressed as a balance quantity at the target item stockroom.

Indented Bill of Material

A multi-level explosion of an assembly or sub-assembly, showing all the levels of inputs, each of which is displayed indented one position from its immediate parent.

Indented Cost Roll Up

A method of simulating the cost of an assembly or sub-assembly with reference to its bill of material and manufacturing operations at all levels, and then rolling up the costs of all its inputs and operations.

Indented Where-Used

This is the inverse of the indented bill of material, and shows the parents of an input, each parent indented one position from its immediate children. The analysis is multi-level, and identifies the parents, grandparents, great grandparents, etc., of an item.

Independent Demand

Demand for an item originating from sales orders or forecasts. That is, direct demand for the item itself.

Ingredient

Any item which is used in the production of another item. See Input.

Input

Refers to any material, sub-component, sub-assembly, ingredient etc. specified on a bill of material/formula/recipe. It is the standard term of reference to any material input.

Input Reference

The key used to access Component Location Reference information. Can also be used as a reference field in its own right. See Component Location Reference.

Input Reference Text

Holds additional text information relating to input references on input items and routes. Used in conjunction with Component Location Reference.

Input Route

The mechanism describing the way that input items are identified and used on bills of materials.

Input Shrinkage

The planned or anticipated percentage of a quantity of material that will be unusable when it is issued to the production process.

Input Where-Used

The identification of where an input is used in assemblies and sub-assemblies.

Inventory Audit Record

When a revaluation of Inventory takes place during a transfer of standard costs from Production, a control record is created for each stockroom revaluation.

Item

The single term used to describe an element of stock identified by a unique number. Also required to identify non-stock or phantom items.

Item Group Minor

Inventory Management classification used in Production Forecasting to define the product family to which an item belongs.

Item Schedule

The planned production of an item expressed as quantities on Due Dates.

Item Stockroom

This is the highest level at which costs and inventory balances are held. The item/stockroom record also defines stock management rules for an item in a stockroom used within Inventory Management.

Item Type

Provides a general classification of an item within the Production system. This may be: made (manufactured/produced); bought out; phantom; reusable tool; consumable tool; gauge or purchased.

Just-In-Time

This is a scheduling and material management philosophy that relies on efficiently organised plants, educated and committed employees, and co-operative suppliers. Its objective is to reduce stock holding to a minimum and optimise the flow of production, synchronised to market demand, thus reducing lead times and increasing customer service.

Key Ingredient

A specific ingredient input on a route that is used to control the lot characteristics of the finished product. Only one key ingredient per route may be defined.

Labour

Work performed by operators.

Labour Time

The length of time required by an operation in terms of labour.

LAD

Acronym for Last Available Date.

Last Available Date

For a lot controlled item, this is equal to the Expiry Date - Customer Shelf Life. It represents the last date on which the item can be used. It is deemed to be frozen after this date.

Latest Cost

One of the Costing Methods available in the Inventory Management application. Using this method, each stock receipt is valued at actual cost and all issues are valued at this cost. In addition, total inventory is valued at this cost.

Lead Time

The amount of time required to produce or procure an item. For production items it is derived from the sum of the lead times of the individual operations required to produce the item and any sub-assemblies. It also relates to procurement times for purchased items. See also Production and Cumulative Lead Times.

Load

The capacity requirement on a work station in terms of time arising from an operation scheduled at that work station.

Location Reference

See Component Location Reference.

Logical Stockroom

A stockroom which does not physically exist but is used as a reference for the recording of WIP inventory, phantom items or floor stock. Recordings may be made to physical stockrooms if they exist, logical stockrooms are simply an alternative.

Lot Balancing Policy

For lot controlled items, an item may be defined such that its potency will determine the actual physical quantity to be issued.

Lot Control

This refers to a level of stock control lower than item/stockroom; also referred to as batch control, for which a group of items received into stock is given a code. Issues from the group require the classification of this code for audit tracking purposes.

Lot Traceability

Where stock control is specified at batch or lot level, this refers to the ability to trace the movement of stock at this detailed level.

Low Level Code

The lowest point in bills of material or production orders at which an item exists. It indicates the maximum level at which the item resides. It is used by MRP to determine when to plan the item in the fully exploded product sequence.

Machine

A piece of equipment upon which or with which work is performed.

Machine Time

The length of time consumed by an operation in terms of machine work.

Master Production Schedule

MPS calculates and balances demand and supply for master scheduled items, and generates a production schedule with suggested dates and quantities.

Material Control Policy

This parameter defines the method of item issues to production. This may be: formal issue; backflush or floor stock issue.

Material Requirements Planning

MRP calculates and balances demand and supply for purchased materials and lower level manufactured items and generates a suggested schedule for production and purchases, with suggested dates and quantities for actions.

Material Type

Parameter used to determine an item’s material type. This may be: direct material; packaging or utility.

Maximum Capacity

The theoretical capacity of a work station in hours when working at its peak rate.

Maximum Capacity Factor

This factor may be applied to a shift profile to allow calculation of the maximum number of hours available at a work station, if for example, the work station consists of several machines or multiple operators. For example, if the work station has a standard shift profile which defines 8 working hours per day, applying a factor of 3 would indicate that 3 x 8 (24) hours are available.

Maximum Order Quantity

A value set for an item to control the suggested supply batch sizes suggested by MPS and MRP. It is an advisory parameter, and does not restrict the size of the suggested batch, but a warning is shown on the plan reports when a batch size exceeds it.

Maximum Stock

The preferred maximum stock balance of an item in a stockroom. This may be set manually for each item

Minimum Order Quantity

A control parameter set for an item to manage the suggested supply batch sizes recommended by MPS and MRP. It ensures that a supply is never less than the defined minimum order value.

Move Days

The length of time required to transport work to a given work station to perform an operation. This is an element of inter-operation time.

Movement Type

This refers to the classification of movements by type of transaction, for example, sundry receipts, customer order issues.

MPS

Acronym for Master Production Scheduling.

MPS Item

An item which is under the scheduling and planning control of Master Production Scheduling. It is typically an end-product, critical sub-assembly, or key material.

MRP

Acronym for Material Requirements Planning.

Multiple Order Quantity

A control parameter set for an item to control the suggested supply batch sizes recommended by MPS and MRP. It defines the increments that are applied to a batch to meet a demand quantity. It sets a defined batch quantity and the ruling that a demand quantity must be supplied in whole batches of the set quantity. For example, demand equals 110; multiple order quantity equals 20; required equals 110/20=5.5 which would convert to 6 batches.

Net Change

An MRP planning method which is driven by exception conditions in the supply and demand status of an item (cf. Regenerative.)

Net Demand

Net demand equals gross demand less available stock, adjusted by demand policy parameters.

Net Requirements

The difference between net demand due on a day and the total suggested supplies planned to be available on that day, adjusted by pre-set Order Policy parameters.

Non-Standard Production Orders

Production orders which are not based on a standard production route, but are created by the user to represent non-standard production operations, resources or input requirements.

On Order

The quantity of an item for which outstanding purchase or production orders exist. It is expressed as a balance quantity at item/stockroom level.

On-Hand Quantity

The quantity shown in Inventory as being physically in stock. For WIP inventory this is calculated as the sum of the Available + Subcontractor + Held balances.

Operation

A stage in the production route of an item

Operation Costs

The costs specific to individual production stages. In the Extended edition of the software, costs can be held at route and operation level as well as item level.

Operational Shrinkage

Percentage loss of work-in-progress as a result of performing an operation.

Order Policy

Order policy is used by MPS and MRP when building a suggested schedule. Policies may be: discrete; discrete above minimum; fixed quantity; number of days supply or multiples above minimum.

Order Release

The point at which a production order is made available for processing on the shop floor. Materials may be allocated and issued at this point.

Order Status

Identifies the stage that a production order has reached. Possible status's are :- suggested, planned, confirmed, released, active, cancelled or completed.

Organisational Model

The organisational model is a control mechanism based on a view of production resources. The model enables the setting of important default values, and the definition of certain procedures and policy issues which will be implemented at resource group level. To use this facility, work stations must be defined to an organisational model.

Output

An item produced as a result of a manufacturing process. This can be a single product, a co-product, by-product, waste or an unplanned product.

Overdue Days (Planning)

Indicates the number of days of overdue supply and demand to be considered in MPS/MRP runs.

Overhead Rate

The rate per hour or % rate applied to absorb production overhead costs in to the item unit cost. Specified on Cost Centres together with an Overhead Recovery Method.

Overhead Recovery Methods

Different recovery methods are available based on production costs, process time, material inputs or outputs in terms of values or quantities.

Overlapped Operations

An operation is defined as an overlapped operation if the next operation can begin before completion of the full quantity at the operation. For example; if 100 items are to be made at operation 10 in batches of 10; but operation 20 can start when 5 batches have been completed at operation 10, then an overlap situation occurs and operation 10 is defined as overlapped. This will be taken into account by planning and scheduling functions.

Overload

The condition where a work station has more work scheduled to be performed than it has available time in a given period.

Parameter File

Contains system and user-defined codes which set control parameters or allow the amendment of standard code descriptions.

Phantom Item

This represents a collection of inputs which are collectively linked together via a 'phantom' item number. This is an item which is not physically stocked but which may be referred to as a generic route input, and will trigger the planning of its component parts via a phantom explosion.

Phantom Operation

A phantom bill of material is provided with a pseudo operation to link its inputs together on a route. This is a phantom operation, and it has no operational impact, although a work station may be assigned to the operation for the purpose of calculating material overheads when the phantom is introduced.

Physical Stock

The total quantity of an item in a stockroom. It is expressed as a balance quantity at item/stockroom level and also at item stockroom lot level.

Pick List

A document detailing the inputs required to be picked for a particular operation on an order or production schedule. This is also referred to as a pulling list.

Planned Available

The quantity calculated to be available at any point in time if MRP or recommendations are implemented.

Planned Down Time

See Down Time.

Planned Material Scrap Rate

This is another way of expressing input shrinkage.

Planned Production Order

A production order that is not yet confirmed, but represents an intention to generate a supply. It does not have input and operation details, and is based on a standard production route.

Planner

A logical grouping of items for the purpose of planning.

Planning Filter

This determines the sensitivity of MPS and MRP rescheduling logic when balancing supply and demand.

Planning Horizon

The end date of an item planning run in MPS or MRP.

Planning Model

A method of defining a view of supply and demand for planning purposes. It is defined in terms of stockrooms. Multiple planning models may be defined to produce differing views of the production environment. One particular model must be defined as that from which MPS or MRP suggestions may be confirmed to production.

Planning Route

This is the route designated for an item to be used in the planning of its input materials and scheduled manufacturing dates and times in MPS and MRP.

Planning Type

The planning category of an item; MPS controlled or MRP controlled.

Potency

A percentage defining the strength of an item in an inventory lot.

Primary Co-Product

The dominant item in a set of process group co-products, which is used to drive the planning for that group of outputs.

Primary Process Group

For a co-product which can be produced in a number of manufacturing process groups, this is the process group to be used as the preferred group in its costing calculation.

Primary Stockroom

The default stockroom for issuing and receipt of an item, when defining a route. On costing routes, the issuing stockroom for an input must be its primary stockroom.

Priority

The relative importance of an order in the work flow. It is used to control the sequence of jobs queuing at work stations.

Process Group Type

Parameter that indicates whether or not the item is a process group in which multiple co-products may be defined.

Process Route

This is a definition of the processes (operational stages) and materials required to produce an item or set of items. It may also be referred to as a production route.

Process Yield

The yield of a process route. It is calculated as the ratio of inputs to the route to outputs from the route.

Product Family

The grouping of related items for forecasting and planning purposes. Group codes are defined on the Inventory Management, Descriptions File, and entered against items in the Inventory Management Product Group Minor field.

Production Calendar

The definition of the production environment in terms of working days, non-working days, holidays and shutdown periods. Production calendars, once defined may be assigned to: Company Profile, Work Stations and MPS/MRP planning profiles.

Production Lead Time

The amount of manufacturing time required to produce an item from its immediate inputs and operations. No reference is made to the lead time of its inputs.

Production Schedule

The plan which contains the sequence and timings of items and operations to achieve the planned production output.

Production Sequence (Major)

An item parameter which controls the sequence in which items are planned in MPS and MRP.

Production Sequence (Minor)

An item parameter which controls the sequence in which item operations are performed, recognising the need to make products in a preferred sequence due to, for example, colour change or set up costs.

Quantity Per

This the standard quantity of a input that is required to make its standard parent lot size.

Quantity Reporting Policy

This policy determines how a WIP inventory quantity booked is interpreted. The quantity recorded may represent the total quantity inclusive or exclusive of scrap and held values.

Queue Time

The length of time that a job will wait, on average, at a work station after arrival before it is worked upon. This is an element of inter-operation time, and should be reduced wherever possible.

Re-Order Point

The quantity of an item in a stockroom which, when reached, should trigger a re-order action. This may be set manually. This Inventory value is used as the safety stock value when using cellular planning. Non-cellular planning, safety stock is taken from the production item master file.

Recipe

See Bill of Material.

Recommended Supply Orders

Suggested replenishments generated by MPS and MRP to support defined inventory stocking policies and to meet outstanding demand.

Regenerative

An MRP planning method which replans every MRP controlled item, regardless of its demand and supply status.

Release Lead Time

This is the time set against a lot controlled item to represent a standard delay between its manufacture/purchase date and its availability for further use/despatch. This lead time is expressed in its Release Lead Time Unit.

Release Lead Time Unit

Indicates the unit in which the Release Lead Time is measured. This may be: days; weeks; months or years.

Released Lead Time Policy

This parameter is pertinent to lot controlled items and allows a set time delay to be taken into account during planning.

Released Production Order

This is a production order which has been released to the production (shop floor) process. Inputs may be allocated and issued to it, and production activities may be booked against it. Any bookings of material or production will automatically change its status to ‘Active’

Repetitive Manufacturing

The style of manufacturing in which high volumes of similar products are produced. Typically, production orders are not used in these environments; but daily production is scheduled against work stations by item and quantity.

Reporting Profile

Although MPS and MRP calculate demand/supply on a daily basis, information pertaining to the production plan may be ‘bucketed’, that is, grouped into time slots, in accordance with a reporting profile defined for each planning model. Usually, this requires the grouping of data into small time periods at the start of the plan; then longer time periods as the plan moves out into future periods.

Reporting Type

On a process route this indicates whether an operation is a count point for WIP inventory; or a backflush (non count) operation. The last operation must be a count point.

They are a part of standard processing rules and transactions which control the effects of booking production.

Resources

These are the facilities which contribute to the production of items. Within the Production system, these comprise: cost centres, work stations, work centres, production calendars, shift profiles, labour grades, operators, crews and subcontractors.

Revision Level

Indicates the current revision level of a route/structure.

Revision Level

Indicates the current revision level of a route/structure.

Rework

This is work necessary to correct a sub-standard item rejected during/after its manufacture.

Rough Cut Capacity Planning

A method of testing the feasibility of an MPS plan by comparing the planned capacity requirements (load) with available capacity at the required production resources at the times required. This may be completed at early planning stages to highlight bottleneck/overload situations before firming/progressing the plan.

Rough Cut Route

The summary bill of capacity used in Rough Cut Capacity Planning. That is, a route/structure which may be set up purely for the purposes of rough cut capacity planning which may be an abridged version of the usual planning route.

Route

A definition of the operational stages involved in producing an item, sequenced in order of manufacture, and specifying the inputs required in terms of materials and resources.

Route Code

The identification code representing an item structure and production method. There can be different routes created for an item. A preferred planning and cost route can be defined.

Route/Structure

This defines both the route (production stages) and material requirements (BOM, recipe, formula) required to produce an item.

Run Time

The length of time required by an operation.

Safety Lead Time (Planning)

Used to set an end date beyond the cumulative lead time of an item. The end date is calculated as item horizon plus safety lead time.

Safety Stock

The desired level of stockholding for an item to support a customer service or availability policy.

Sales Forecast

A statement of the anticipated market demand for a product. It can be compared with actual sales orders, in MPS/MRP calculations to determine the net demand to be met by production. This is dependent upon the Demand Policy code set for the item.

Schedule

See Production Schedule.

Schedule Control

An environment in which item/work station schedules are used in preference to production orders - usually in a high volume, repetitive form of production.

Schedule Controlled Item

An item that is schedule and not production order controlled in MPS and MRP processes. A production order can be raised if required.

Scheduled Receipt

A planned supply in MPS/MRP; this may be a released or active production or purchase order or a suggested or confirmed schedule.

Scheduling

The process of calculating and suggesting due dates, quantities and action dates for the supply of an item to meet required demand quantities and dates.

Seasonal Profile

Method used to spread forecasts using indices for each forecast period and entering a total figure to spread. Can be used to speedily determine forecast values which display seasonal fluctuations.

Serial Number Control

A form of lot control which maintains single, uniquely identified (serialised) units.

Set Up

The activity of preparing machines or processes for production. Set up time forms part of the lead time of an operation.

Set Up Time

The duration of the set up for a work station. This is expressed as a labour time.

Shelf Life

The life of an item expressed in its Shelf Life Unit.

Shelf Life Unit

Indicates the unit in which an item’s shelf life is measured. this may be: days; weeks; months; years or unlimited.

Shift Length

The duration of an individual working shift for a work station.

Shift Profiles

These describe the pattern of shifts in a day. Shift profiles use effectivity dates to reflect planned changes in patterns. A default shift profile may be assigned to a work station; or a shift profile assigned to each working day within a week at a work station. The shift profile defines the number of productive hours available on a working day.

Shipper Number

A number assigned to each shipment of items to or from a subcontractor if Shipper Tracking is in use.

Shipper Tracking

A method of tracking materials or WIP inventory to or from subcontractors.

Shrinkage (Material)

The planning factor applied to an input on a route to reflect expected loss.

Shrinkage (Operation)

The planning factor applied to an operation to reflect expected losses. Scheduling uses the factor to inflate the standard times to make the required lot size.

Shrinkage Cost

The amount of item unit cost attributable to operational or material shrinkage in the production process. It is held by Cost Element and can optionally be consolidated into the item cost elements. A shrinkage element can be configured to display the total shrinkage cost.

Simulated Cost

A function which projects product costs by applying variables to the cost structure to ascertain likely future costs, or by changing inputs to ascertain the cost impact of the changes.

Single Level Enquiry

A one level explosion of a bill of material and route and which costs the inputs and operation processes required to make the parent item.

Smoothing Policy

A planning policy which smoothes sale forecast demand to provide a level production schedule.

Specification Ref

This refers to how an item is pecified.

Standard Capacity

The daily capacity in hours of a work station when operating at its normal rate, and normal shift patterns.

Standard Capacity Factor

This may be applied to a shift profile to determine the standard number of hours available at a work station. In situations where the work station comprises multiple machines or personnel, the factor will represent the number of machines/people at that work station. For example, for a shift profile of 10 hours at a work station where 2 machines operate, a capacity factor of 2 would be entered, to indicate a standard capacity of 20 hours.

Standard Costs

A costing method available in Production and Inventory. Standard costs are calculated for items based on standard cost rates and operation times and the standard costs of inputs. They form the yardstick for performance measurement in a given period.

Standard Efficiency

The percentage of the standard capacity of a work station which you expect to achieve under normal operational circumstances. This percentage may be used in capacity planning enquiries and reports.

Standard Lot Size

Standard batch size in terms of which input quantities and operation times are expressed in a route/structure.

Standard Potency

The standard strength of an item expressed as a percentage. Applies to lot controlled items only.

Standard Production Orders

Production orders which are based on a standard route to obtain input requirements and operation details.

Start Date

The scheduled release date of a production or purchase order or schedule.

Start Date (Planning)

This is the first date considered by MPS and MRP Demand and Supply prior to this date is ignored. It is the Current Date less Overdue days set for the planning run.

Stock Forecast

A forecast used in MPS and MRP to plan variable levels of inventory availability to maintain desired customer service levels over and above standard safety stock

Stock Monitor

An Inventory Management function which maintains the integrity of lot controlled stock availability. It determines whether a lot is available or has passed its Last Available or Expiry date. All lots are frozen when the Last Available Date is passed.

Stock Movement

A movement of a quantity of an item into or out of a stockroom. More particularly, this will refer to the recording of such a movement in the application, and the transaction record created as a result.

Stock Run Out Policy

Controls the planning of requirements of an item based on its stock balance, rather than effective dates. The available policies are: use up stock and do not replan; or use up stock and then use a nominated replacement item or items.

Stockroom

A discrete area where stock for an item is recorded and controlled separately from other company stocks. Stockroom codes are also used to define ‘logical’ stockrooms used to hold WIP inventory, and Floor Stock Locations which may be physical or logical stockrooms.

Stockroom Balance

See Balance and Item/Stockroom.

Subcontract Operation

Work on the production of an item which is carried out by another manufacturer. This entails sending materials or WIP which are worked on by the subcontractor before being returned for further operations, or quality inspection or receipt into stock.

Subcontractor Stockroom

A logical stockroom which holds all subcontractor *material* balances. Subcontractor *WIP inventory* balances are held as balances at operations in the associated work station WIP location.

Substitute

An item which has been designated as an allowable replacement for an item. It may be issued in whole or part to a production order, if there is insufficient stock of the primary item.

Substitution Policy

Defined on a route/structure input item definition, indicating whether it is permissible to use a substitute item if there is a stock shortage of the primary item.

Suggested Production Order

An MPS or MRP recommendation to create a production order to satisfy a shortage identified by the planning process.

Suggested Purchase

An MPS or MRP recommendation to create a purchase order to satisfy a shortage identified by the planning process.

Supply

The planned or scheduled receipt of item quantity from a purchase order or production order or a production schedule item.

Target Yield

Desired yield of a route.

This Level

The final level of manufacture for an item with a multi level route/structure, as opposed to ‘lower’ levels of manufacture such as sub-assemblies.

Time Basis Code

Code indicating how operation times are expressed on a route. These are: time per lot; time each; quantity per hour; fixed time, time per 1000; time per 100; time per fixed batch.

Time Booking Policy

Parameter set on the Organisational Model to control the time booking format in Production reporting. This may be in decimal hours or hours and minutes. This policy is set only if the Time Reporting Policy is set to elapsed time.

Time Fence

The date which the schedule is fixed no recommendations are made by MPS or MRP to change existing production or to suggest new production.

Time Fence Days (Planning)

The number of days that are added to the Current Date to calculate the Time Fence Date.

Time Fence Policy

Parameter set at item level indicating whether shortages occurring within the time fence should be ignored, or satisfied on the Time Fence date.

Time Reporting Policy

Parameter set on the Organisational Model to control the format in which operator and work station times at an operation are entered. This may be set for entry as elapsed time or as work start time and stop time.

Time Units

The units in which operation times are expressed. This is defined in the Company Profile and can be in hours or minutes.

Total Shelf Life

The life of an item lot. The shelf life is added to the creation date to calculate the Expiry Date.

Transaction Manager

The function that processes production and WIP inventory transactions, generating movement records and updates balances. It runs in its own subsystem and may be started and stopped. It must be running in order to keep balances and transaction details up-to-date during production bookings.

Transaction Number

Each production booking entered on the system is allocated a system transaction number which may be accessed and displayed for subsequent reference in enquiries and reports.

Transaction Type

JBA transaction codes which represent a particular balance update or movement generation. The transaction type calls a program which ultimately updates the database.

Trial Kit

A method of simulating input allocation to a production order or route to assess availability to meet the requirements. Also known as Material Availability Enquiry.

Trigger

The mechanism used to drive Net Change MRP. Item Triggers are created when transactions are recorded for ‘unplanned’ events. Triggers may be generated through maintenance changes; sales/purchase/production orders set up changes; stock issues/receipts; MPS/MRP schedule amendments.

Trigger Tolerance

This is the percentage (above or below) of safety stock which, if breached by the projected available stock, will cause a net change trigger to be written for the item.

Unit Cost

The amortised cost of a single unit of an item.

Unit of Measure

The unit in which a balance quantity or unit cost is expressed.

Unplanned Issue

Issue of inputs to a production order which has not been previously allocated.

Unplanned Receipt

Receipt into inventory of an item or items not expected at the booking operation. That is, not standard on the route, or order.

Usage

The quantity of an item issued from a stockroom in a given period.

Usage Profile

A user-defined profile which specifies the pattern of periods to be included in the calculation of average usage.

Utilisation

The extent to which the capacity of a work station is expended by actual work performed.

Utility

An item may be defined with a Material Type of utility. For example, electricity, gas, water etc.. Items of this type do not have a physical balance changed when used in production.

Value/Usage

The value/usage setting for an item in Inventory. It positions the item in a matrix of value/usage. It is a selection criterion for selective MRP.

Variance

A difference between the standard cost or volume of a process and the actual recorded cost or volume.

Waste Product

An output from a process route which does not have any intrinsic worth or saleable value; and which may incur a cost in its disposal or shortage.

WIP

Acronym for Work-In-Progress. Also known as Work-In-Process.

WIP Inventory

Work-in-progress inventory; transparent to Inventory Management, but accessible through enquiries in Production WIP Inventory Control.

WIP Location

A WIP location is a stockroom that has been 'logically' associated with one or more work stations as the stockroom to hold WIP inventory balances produced at count point operations.

Work Centre

A collection of work stations that have been grouped together for capacity requirements analysis purposes. Work centres are not used in planning or work station scheduling.

Work Station

The standard production unit or facility for which capacity requirements are measured.

Work Station Schedule

A daily work plan for a work station, containing item and order quantities and duration of set up and operating hours.

Work-In-Progress

The value of work currently underway in the factory in terms of the material issued, and the operations performed. For a given order or schedule, it is calculated as the value of material and work input less the value of receipts made into stock. Work-in-progress (WIP) can be valued at standard or current cost. Also known as work-in-progress.

Yield Item

An item that is sensitive to yield either as an input or an output. Yield is the ratio of total quantity of outputs compared to the total quantity of inputs.

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